FINAL YEAR PROJECT PROPOSAL

AI-Based Emergency Vehicle Priority and Dynamic Green Corridor System for Intelligent Traffic Management

AI-Driven Emergency Traffic Management Using Real-Time Video Analytics, Traffic Prediction and Dynamic Multi-Intersection Signal Control

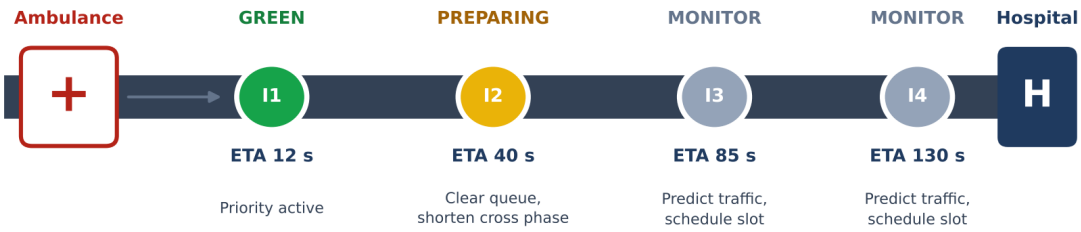


Figure 1: The moving green corridor – intersections are prepared ahead of the ambulance and restored behind it.

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Institution	University of Management and Technology (UMT), Lahore Computer Science – School of System & Technology
Proposed environment	Punjab Safe Cities Authority (PSCA) – PPIC3, Lahore (proposed only; not yet confirmed)
Project type	Final Year Project (Artificial Intelligence / Intelligent Transportation Systems)
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Abstract

This project proposes an AI decision layer that gives ambulances a predictive, safety-validated green corridor across a whole route, instead of one intersection at a time. Ambulances lose critical minutes at congested signalised intersections, and green corridors are still largely arranged by hand.

Given an emergency request, the system selects a traffic-aware route to the most suitable hospital, predicts when the ambulance will reach each intersection, and coordinates a moving green corridor, restoring normal signal operation behind the vehicle. Development and evaluation run in a SUMO-based digital twin of a Lahore corridor. The architecture is designed to plug into the live data and signal infrastructure of the Punjab Safe Cities Authority (PSCA) wherever access is later authorised; no PSCA collaboration is confirmed at this stage (Section 14).

1. Project Team and Supervision

The team has three members; the project is supervised by two UMT faculty members. A PSCA technical mentor has not been confirmed.

1.1 Team members

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1	Muhammad Ahmad Asif	F2023266537	F2023266537@umt.edu.pk
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1.2 Supervision

Role	Name	Email
Project advisor	Prof. Hafiz Ahsan Arshad	ahsan.arshad@umt.edu.pk
Project co-advisor	Dr. Muhammad Husnain Ashfaq	husnain.ashfaq@umt.edu.pk
PSCA technical mentor	Not yet confirmed	N/A

Institution: University of Management and Technology (UMT), Lahore. Department: Computer Science, School of System & Technology.

1.3 Proposed work distribution

The project is divided into three workstreams so each member owns a clear area, while all members contribute to evaluation and the final thesis. Ownership will be finalised with the advisor and co-advisor.

Workstream	Scope
A – Data, Perception and Prediction	PSCA data requirements, GIS / PostGIS, computer vision (YOLO + tracking), traffic analytics, ETA and traffic-prediction models.
B – Simulation, Routing and Control	SUMO digital twin, TraCI / libsumo, traffic-aware A* routing, green-corridor engine, OR-Tools optimisation, PPO controller, safety layer.
C – Platform, Integration and Evaluation	FastAPI backend, database schema, data-provider and signal-controller adapters, React dashboard, agentic layer, testing, experiments, documentation.

2. Problem Statement

Ambulances in Lahore cross many congested signalised intersections with no automated, route-wide priority, and every minute of delay reduces a patient's chance of a good outcome.

2.1 Background

Lahore is Pakistan's second-largest city and carries dense, mixed traffic: cars, motorcycles, rickshaws, buses and trucks share the same signalised corridors, and peak-hour congestion is routine. An ambulance responding to a cardiac arrest, stroke or major trauma must cross many of these intersections.

In practice, ambulances rely on sirens, the driver's judgement and occasionally manual coordination with traffic staff. A green corridor, when arranged at all, is normally set up by radio or phone, covers a handful of intersections and depends on human operators.

The Punjab Police Integrated Command, Control and Communication Centre (PPIC3) in Lahore already brings together intelligent traffic management, a large CCTV network, centralised signal control (PTSC), GPS-based dispatch (AVLS) and 1122 emergency coordination. What public documentation does not show is an automated decision layer that uses these to serve a specific emergency vehicle.

2.2 The gaps

- **Reactive, not predictive.** Priority is triggered only when the ambulance is close to a signal, leaving no time to clear the queue already formed.
- **One intersection at a time.** Existing preemption acts locally; nothing plans arrival time at every intersection along the route.
- **Shortest distance is not fastest time.** Routes ignore predicted congestion, signal delay and incidents.
- **Manual and hard to scale.** Operators cannot hand-configure corridors for several simultaneous emergencies.
- **Side-effects are ignored.** Turning every signal green causes cross-traffic gridlock and long recovery times.
- **No safe place to experiment.** Live signals cannot be a test bench, so new strategies are hard to evaluate before deployment.

2.3 Formal problem statement

There is no automated, predictive and safety-validated decision layer that, for a given emergency vehicle, (i) selects a traffic-aware route to the most suitable hospital, (ii) forecasts when the vehicle will reach each intersection on that route, and (iii) coordinates signals across the entire route, while limiting delay to normal traffic and preventing unsafe signal states.

Our contribution is emergency-specific, predictive, multi-intersection coordination, not "AI traffic management" in general. Verifying what PSCA's ITMS already provides is the first task of the project (Section 12).

3. Proposed Solution

We propose an AI emergency-priority decision layer built on top of existing traffic and emergency-response infrastructure. It does not replace cameras, ITMS or signal controllers: it consumes their data, decides what should happen along an ambulance's route, validates every action for safety, and issues the resulting signal plan.

3.1 How the system works

When an emergency request arrives, the following pipeline runs automatically, without an operator opening signals one by one:

1. **Emergency request received** – incident location, priority and assigned ambulance.
2. **Traffic state acquired** – vehicle count, speed, density and congestion level from CCTV / ITMS data.
3. **Ambulance located** – GPS position, speed and heading from AVLS, map-matched to the road network.
4. **Hospital selected** – the most suitable destination by predicted travel time (and capacity or speciality data, if available).
5. **Route computed** – traffic-aware A* using predicted travel time, signal delay and incident penalties rather than distance.
6. **Arrival predicted** – estimated arrival at the destination and at every upcoming intersection.
7. **Intersection traffic forecast** – expected queues and flow at each intersection when the ambulance will arrive.

8. **Green corridor planned** – which signals, when to start preparing, and how long to hold priority.
9. **Safety validation** – every command is checked for conflicts, minimum green, pedestrian phases and cross-traffic impact.
10. **Priority executed** – commands are applied in SUMO and, only where PSCA authorises, requested through the PTSC interface.
11. **Progress monitored** – route and corridor are re-planned if traffic changes (for example an accident ahead).
12. **Normal operation restored** – each intersection returns to its regular plan as soon as the ambulance has cleared it.

3.2 What makes the solution intelligent

- **Predictive control.** Signals are prepared while the ambulance is still, say, 40 seconds away, so queues clear before it arrives.
- **Route-wide coordination.** One plan covers all intersections on the route and moves with the vehicle.
- **Traffic-aware routing.** Edge costs use predicted travel time, congestion and incident penalties, so the chosen route can differ from the shortest one.
- **Multi-objective decisions.** The controller minimises ambulance delay while limiting extra delay, queue growth and signal switching for normal traffic.
- **Dynamic re-routing.** If congestion or an incident changes the best route, route and corridor are recomputed on the fly.
- **Safe by design.** AI components only recommend; a rule-based safety layer approves or rejects every signal command.

4. Existing Systems and How Our Project Differs

Several systems address parts of this problem, but none combines traffic-aware routing, per-intersection arrival prediction, route-wide corridor planning and independent safety validation in one decision layer.

4.1 Existing approaches

Existing approach	What it does	Limitation relative to our goal
Manual green corridors (police / operators)	Operators coordinate signals along a route by radio or phone.	Labour-intensive, covers few intersections, no prediction, does not scale to concurrent emergencies.
Opticom-type preemption (3M / GTT / Miovision)	On-vehicle unit sends a request to equipped intersections; controller gives green on the ambulance's approach, then resumes normal operation.	Works intersection by intersection and needs compatible equipment on vehicle and signal. Does not choose route or hospital, forecast downstream traffic or plan the corridor as a whole.
SCATS route preemption	Adaptive system whose route preemption facility lets agencies define emergency plans for routes.	Emergency routes are pre-defined rather than computed live; no per-signal arrival forecast or live traffic-aware hospital and route choice.
Adaptive control (SCOOT, SCATS, SURTRAC)	Continuously adapts cycle, split and offset to observed traffic.	Optimised for normal traffic; literature notes slower response to abnormal situations such as incidents and emergencies.
Navigation apps (Google Maps, Waze)	Live traffic-aware routing for drivers.	Cannot control signals, give emergency priority or select hospitals.
PSCA ITMS / PTSC (existing platform)	CCTV analytics, journey-time monitoring, central signal control, dispatch and 1122 coordination.	The foundation we build on, not a competitor. Public documentation does not describe an automated, predictive, route-wide emergency corridor; to be confirmed with PSCA.
Academic RL / optimisation studies	Reinforcement learning or optimisation for emergency-vehicle signal control.	Typically single-intersection or synthetic networks, rarely combined with live routing, prediction, a safety layer and real city data.

4.2 Capability comparison

Capability	Manual corridor	Opticom-type EV	SCATS route pre-emption	Adaptive control	Navigation apps	Proposed system
Traffic-aware route to hospital	Partial	No	Partial	No	Yes	Yes
Predicts arrival at each intersection	No	Partial	No	No	Partial	Yes
Coordinates the whole route	Partial	No	Yes	Partial	No	Yes
Considers downstream congestion and cross-traffic impact	Partial	No	No	Partial	No	Yes
Independent safety validation of each command	Partial	Partial	Partial	Partial	No	Yes
No new vehicle or roadside hardware	Yes	No	Partial	Partial	Yes	Yes*
Tested in a digital twin before deployment	No	No	No	No	No	Yes

* Relies on AVLS / GPS data, subject to PSCA authorisation. Ratings are our assessment from public descriptions and vary by vendor version and deployment.

4.3 Why our project is different

1. **Predictive, not reactive.** We forecast ambulance arrival and intersection traffic and prepare signals ahead of time.
2. **Routing and signals in one loop.** Hospital and route selection, corridor planning and signal control share the same digital twin and predictions.
3. **Aware of side-effects.** The objective penalises delay to normal traffic, queue growth and signal switching, so the corridor does not create gridlock elsewhere.
4. **Safety layer between AI and signals.** Machine-learning and LLM components never command a signal directly; a rule-based validator does.
5. **Built for PSCA's ecosystem.** It reuses existing data (CCTV analytics, AVLS, PTSC) instead of new roadside hardware, and is designed for Lahore's mixed traffic, including motorcycles and rickshaws.
6. **Measurable and reproducible.** A calibrated SUMO digital twin lets us compare six control strategies on the same scenarios with quantitative metrics.

4.4 Public sources consulted

- Miovision, Opticom emergency vehicle preemption overview – www2.miovision.com/opticom
- R. Baskerville, "Adelaide's high tech – low cost emergency vehicle priority system" (SCATS route pre-emption) – swov.nl
- Sydney Coordinated Adaptive Traffic System (SCATS) overview – en.wikipedia.org/wiki/Sydney_Coordinated_Adaptive_Traffic_System
- Incorporating V2I communications into adaptive traffic control systems – arXiv:1606.01010
- Punjab Police, PPIC3 overview – lahorepolice.punjab.gov.pk/ppic3; Punjab Safe Cities Authority – punjabpolice.gov.pk/psca

5. Project Overview

The project delivers a complete, evaluated emergency-priority layer, proven first in simulation and designed to connect to PSCA infrastructure only if authorised.

5.1 Aim

To design, build and evaluate an AI-based emergency traffic-management layer that reduces ambulance travel time by combining real-time traffic understanding, prediction, traffic-aware routing and safe, coordinated multi-intersection signal priority.

5.2 Objectives

1. Establish, with PSCA (if collaboration is confirmed), exactly which traffic, camera, signal, GIS and emergency data can be used, and under what security conditions.
2. Build a GIS-based digital twin of a real Lahore corridor in SUMO, calibrated with available traffic observations.
3. Extract traffic state (count, speed, density, queue, congestion level) from video or from processed detections.
4. Predict ambulance ETA and short-term traffic (speed, volume, travel time, congestion) at 1, 3 and 5 minutes.
5. Implement traffic-aware routing and hospital selection with dynamic re-routing.
6. Implement a predictive, multi-intersection green-corridor engine with an independent safety layer.
7. Investigate a reinforcement-learning (PPO) controller for multi-objective signal optimisation.
8. Deliver an operator dashboard and quantitatively evaluate the system against baseline strategies.

5.3 Scope

In scope	Out of scope
Digital twin of a selected Lahore corridor	Replacing PSCA's ITMS, cameras or signal hardware
Computer-vision traffic analytics	Installing new roadside or on-vehicle hardware
ETA and traffic prediction	Uncontrolled commands to live signals
Traffic-aware routing and hospital selection	City-wide deployment
Green-corridor planning and safety validation	Fine-tuning large language models
RL controller and constrained agentic layer	Public-cloud-dependent architecture
Operator dashboard	
Adapters for PSCA data and PTSC (used only if authorised)	

5.4 Operating modes and data adapters

Mode	How it works	Purpose
Mode 1 – PSCA Real Data	Authorised PSCA data (traffic, ambulance location, signal state, GIS) flows through the same AI pipeline and produces live recommendations. Available only if PSCA collaboration is confirmed.	Real-world validation and integration.
Mode 2 – Digital Twin / Simulation	Traffic observations (PSCA-provided if available, otherwise public or synthetic data) calibrate a SUMO model of a Lahore corridor; the AI controller acts on simulated signals through TraCI.	Main development, training and evaluation environment; works even if no PSCA access is granted.

To keep the AI independent of where data comes from, the code uses two interfaces: a TrafficDataProvider (PSCA, simulation or test data) and a SignalController (SUMO or PSCA). Switching modes means swapping an adapter, not rewriting the system.

6. Functional and Non-Functional Requirements

This section consolidates what the system must do (functional) and the quality and operating constraints it must satisfy (non-functional).

6.1 Functional requirements

ID	The system must…
FR1	Accept an emergency request containing incident location, priority and assigned ambulance.
FR2	Ingest traffic state (count, speed, density, queue, congestion level) from video, processed detections or simulation.

ID	The system must…
FR3	Locate the ambulance from GPS data and map-match it to the road network.
FR4	Select the most suitable hospital and compute a traffic-aware route to it by predicted travel time.
FR5	Predict ambulance ETA at the destination and at every upcoming intersection.
FR6	Forecast short-term traffic (speed, volume, travel time, congestion) at 1, 3 and 5 minutes.
FR7	Plan a multi-intersection green corridor: which signals, when to prepare, how long to hold priority.
FR8	Validate every signal command for conflicts, minimum green, pedestrian phases and cross-traffic impact before execution.
FR9	Apply validated commands to SUMO signals and, only where authorised, to the PSCA PTSC interface.
FR10	Re-plan route and corridor when conditions change (for example an incident) and restore each intersection to its normal plan after the ambulance passes.
FR11	Display a live operator dashboard: map, route, ETA, signal states, AI decisions, time saved versus baseline and metrics.
FR12	Log all decisions, including rejected commands, and support role-based access (Admin, PSCA Operator, Traffic Operator, Research read-only).

6.2 Non-functional requirements

ID	Category	Requirement
NFR1	Safety	AI components only recommend; a rule-based validator approves or rejects every command, and no LLM issues signal commands.
NFR2	Performance	Route, ETA and corridor decisions are produced fast enough to be useful in real time; the dashboard updates every one to two seconds via WebSocket. Target latencies will be fixed after the first prototype is measured.
NFR3	Accuracy	Prediction models are measured on held-out data with MAE, RMSE and classification accuracy, and compared with simple baselines.
NFR4	Reliability	The system falls back safely to normal signal operation if data is missing, a module fails or a command is rejected.
NFR5	Security and privacy	Sensitive data is processed locally, vehicles use anonymous IDs, access is role-based and audit logs are kept.
NFR6	Deployability	Offline-first and on-premise; no dependency on a public cloud service.
NFR7	Modularity	Data sources and signal control sit behind adapter interfaces, so switching between simulation and PSCA data means swapping an adapter.
NFR8	Reproducibility	Experiments run on identical SUMO scenarios with repeated random seeds and tracked parameters.
NFR9	Usability	Operators can see and understand each AI decision and the corridor status at a glance.
NFR10	Maintainability	Open-source tools, versioned code, automated tests and documentation, especially for the safety layer.
NFR11	Resource limits	Runs with pretrained models and CPU inference where GPU or software access is restricted.

7. Core AI Functionalities

The project rests on three core AI functionalities, built in this order, plus two advanced enhancements added only after the core corridor works.

7.1 Core AI functionalities

ID	AI functionality	Technique	What it does	Obj.
AI-1	Computer-vision traffic perception	Fine-tuned YOLO detector, ByteTrack / BoT-SORT tracking, homography calibration	Detects and tracks cars, buses, trucks, motorcycles and ambulances; derives count, flow, speed, density, queue length and congestion class.	3

ID	AI functionality	Technique	What it does	Obj.
AI-2	Machine-learning ETA and traffic prediction	Linear / random-forest baselines, XGBoost / LightGBM, then LSTM / GRU	Predicts ambulance ETA and short-term speed, volume, travel time and congestion.	4
AI-3	Traffic-aware routing and hospital selection	Time-dependent, traffic-weighted A* (after Dijkstra and A* baselines)	Ranks hospitals by predicted arrival time and returns the best hospital-and-route pair, with dynamic re-routing.	5

7.2 Advanced AI enhancements

ID	AI functionality	Technique	What it does	Obj.
AI-4	Reinforcement-learning signal control	PPO in a Gymnasium environment wrapped around SUMO, plus OR-Tools constrained optimisation	Investigates multi-objective signal control that reduces ambulance delay without excessive delay to normal traffic.	6, 7
AI-5	Constrained agentic layer (optional)	LangGraph Emergency Manager with Traffic, Route and Signal agents, local LLM	Produces a high-level plan and operator-facing explanations; never issues signal commands.	6

All AI outputs pass through the rule-based safety layer before any signal command is produced.

8. System Architecture and Core Modules

The architecture is layered, offline-first and modular: data flows from authorised sources into a digital twin, through the AI engine and corridor planner, and through the safety layer before any signal command is produced.

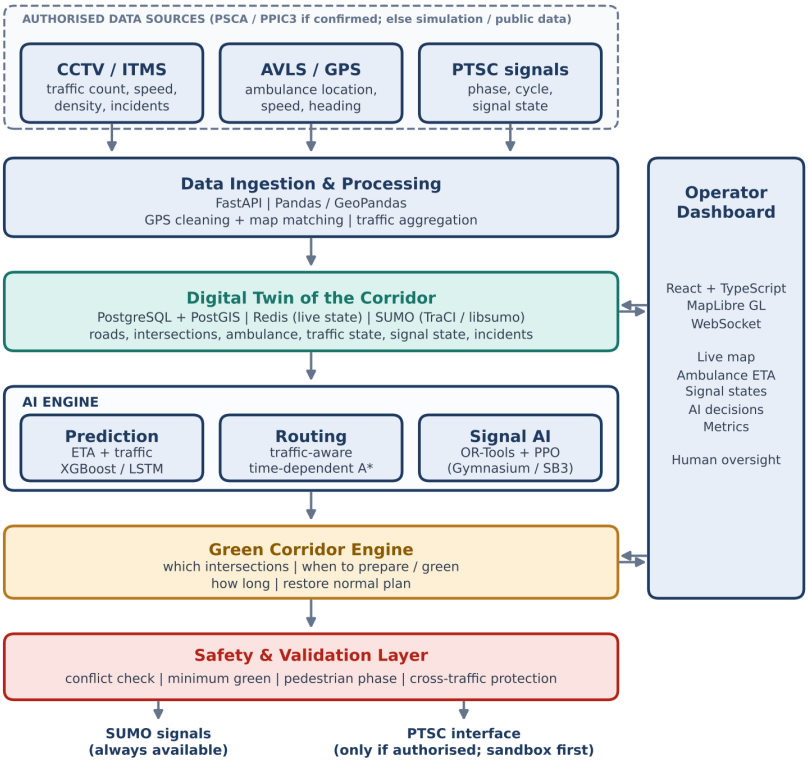


Figure 2: System architecture from data sources to validated signal commands.

8.1 Architecture layers

Layer	Responsibility
Data sources	CCTV / ITMS traffic information, AVLS ambulance location, PTSC signal state, GIS road and intersection data (all subject to PSCA authorisation; simulation and public data otherwise).

Layer	Responsibility
Ingestion and processing	Secure API or feed intake, validation, GPS cleaning and map matching, traffic aggregation (FastAPI, Pandas / GeoPandas).
Digital twin	Live representation of the corridor: road graph, ambulance position, traffic state, signal state and incidents, mirrored in SUMO (PostgreSQL + PostGIS, Redis).
AI engine	ETA and traffic prediction, route optimisation, signal optimisation.
Green-corridor engine	Chooses intersections, timing and duration of priority and the restore plan.
Safety and validation	Rule-based checks on every command; rejects anything unsafe.
Control layer	SUMO signals (always) and PTSC interface (only if authorised, sandbox or read-only first).
Dashboard	Live map, ETA, signal states, AI decisions, metrics and operator oversight (React).

8.2 Core modules

A. Perception and traffic-state estimation. A pretrained YOLO detector is fine-tuned on approved or recorded footage; ByteTrack or BoT-SORT keeps vehicle IDs across frames; homography converts pixel movement into metres. If only processed detections are available, the same downstream modules consume them directly.

B. Prediction. Models grow in complexity: linear and random-forest baselines, then XGBoost / LightGBM, then LSTM / GRU where enough time-series data exists. Targets are ambulance ETA, volume, speed, travel time and congestion level.

C. Routing and hospital selection. A directed road graph whose edge cost is predicted travel time plus congestion, incident and signal-delay penalties. Dijkstra and A* come first, then a time-dependent traffic-weighted A*.

D. Green-corridor engine. Once a route is fixed, the engine calculates expected arrival at every intersection and schedules signal actions ahead of it, following emergency vehicle preemption (EVP) principles.

Intersection	Ambulance ETA	Action
I1	35 s	Green priority now; hold until the ambulance has cleared.
I2	70 s	Prepare: predict queue, shorten cross-street phase, clear approach.
I3	110 s	Monitor and reserve a green slot.
I4	155 s	Monitor and reserve a green slot.

As the ambulance passes I1 it returns to normal operation, I2 goes green, and I3 and I4 move to preparation: a corridor that travels with the vehicle.

E. Signal optimisation and reinforcement learning. OR-Tools constrained optimisation is the first controller. A PPO agent trained in a Gymnasium environment around SUMO (via libsumo) is then investigated.

Element	Design
State	Lane densities and queues, current phase and duration, ambulance distance and ETA, downstream density.
Actions	Keep phase, change phase, extend green, apply emergency priority (final set follows the actual signal configuration).
Reward	Negative ambulance delay, normal-traffic delay, queue length and unnecessary phase switching; positive emergency progress. A reward that only rewards ambulance movement is rejected because it can ruin all other traffic.

F. Safety and validation layer. Every command is checked for valid phase, minimum green and maximum red, no conflicting greens, protected pedestrian phases, cross-traffic impact within limits and permitted emergency override. Rejected commands are logged; in live use, actions would also require operator authorisation.

G. Constrained agentic layer. A LangGraph Emergency Manager coordinates Traffic, Route and Signal agents. The language model produces a high-level plan; optimisation or RL computes timings; the safety layer validates them. It is an enhancement added after the core corridor works.

H. Operator dashboard. A React dashboard shows the ambulance, route and intersections on a live map, traffic state, signal status (green / preparing / normal), ETA, time saved versus baseline, AI decisions and system metrics.

9. Research Questions and Evaluation Plan

Six research questions are answered by comparing six control strategies on identical SUMO scenarios.

9.1 Research questions

ID	Question
RQ1	Can traffic observations be used to build a sufficiently accurate traffic digital twin?
RQ2	Does traffic-aware emergency routing reduce ambulance travel time compared with shortest-distance routing?
RQ3	Does coordinated multi-intersection preemption reduce emergency waiting time?
RQ4	Does predictive signal preparation outperform reactive preemption?
RQ5	Can an RL controller reduce emergency delay without excessive extra delay to normal traffic?
RQ6	Can the system recover and re-route when traffic conditions change (for example an incident)?

9.2 Evaluation plan

Strategies are compared on off-peak, peak and incident scenarios, with repeated runs across random seeds.

ID	Strategy
A	Normal signals (baseline)
B	Traditional reactive emergency preemption
C	Traffic-aware routing only
D	Traffic-aware routing + green corridor
E	Prediction + green corridor
F	PPO + prediction + green corridor

Metrics: ambulance travel time, ambulance waiting time, number of ambulance stops, normal-traffic delay, queue length, throughput, signal-switching count and recovery time. Prediction models are additionally evaluated with MAE, RMSE and classification accuracy on held-out data.

10. Timeline, Assumptions and Risks

The plan runs about six months and will be aligned with the university schedule. The core FYP is complete at the end of Phase 4 (Month 3); Phases 5 and 6 are enhancements, and Phase 7 covers evaluation and reporting.

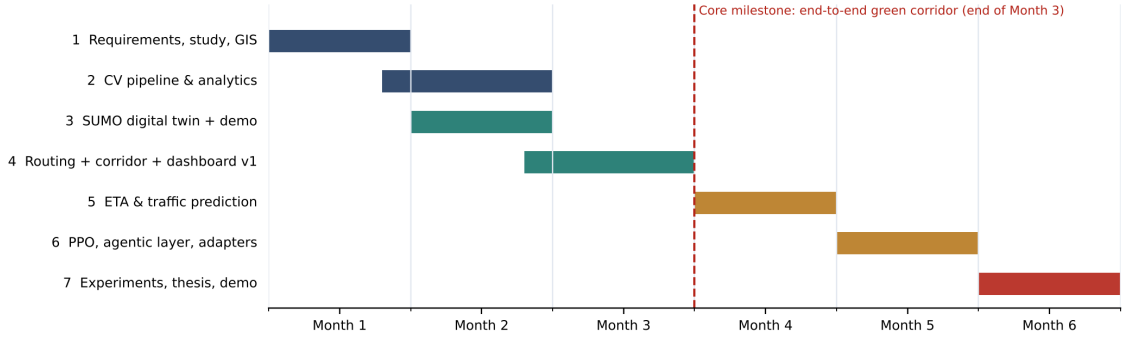


Figure 3: Six-month development timeline.

10.1 Development phases

Phase	Timeline	Activities	Milestone
1	Month 1	Domain study (traffic engineering, EVP, digital twins); requirements definition (with PSCA if confirmed); environment setup; GIS / PostGIS and data preparation.	Technical requirements document
2	Months 1–2	Computer-vision pipeline; traffic analytics (detection, tracking, speed, density).	Vehicle counting, tracking and congestion levels
3	Month 2	SUMO network of a Lahore corridor; TraCI control; one intersection + one ambulance + one signal.	First demo: ambulance receives priority
4	Months 2–3	Traffic-aware A*, hospital selection, multi-intersection corridor, safety layer, dashboard v1.	Core milestone: end-to-end green corridor
5	Month 4	ETA and traffic prediction (XGBoost, then LSTM).	Predictive corridor
6	Month 5	Gymnasium environment; PPO training; comparison with rule-based control; optional agentic layer; PSCA adapters; dashboard completion.	RL controller and integrated system
7	Month 6	Experiments A–F, analysis, documentation, thesis, final demonstration.	Final report and demo

10.2 Assumptions and risks

Risk / assumption	Mitigation
PSCA collaboration or data access is not confirmed or not granted	The system is fully functional in SUMO; public, historical, processed or synthetic data and adapters make PSCA data an add-on, not a prerequisite.
No permission for direct signal control	SUMO is the control target; commands are generated and validated, and demonstrated in a sandbox or read-only mode.
Only processed detections, no raw video	The pipeline accepts counts and speeds directly; the CV module runs on approved or recorded footage only.
Limited GPU or restricted software	Use pretrained models with fine-tuning, smaller architectures and CPU inference; keep the stack offline-first.
Simulation-to-reality gap	Calibrate SUMO with available observations, run sensitivity analysis and state the limitation explicitly.
RL instability or unsafe policies	Start with rule-based control; RL acts only through the safety layer; multi-objective reward.
Scope creep	Milestone-gated plan; prediction, RL and agentic AI are enhancements after the core corridor works.

11. Target Users and Beneficiaries

The system is designed for emergency-response and traffic-management staff, and benefits patients and the wider public through faster ambulance arrival. PSCA is the intended operating environment, not a confirmed partner.

Group	Role	How they benefit
PSCA / PPIC3 operators and traffic-management personnel	Primary users of the dashboard	See route, ETA and corridor status in one view; no manual signal-by-signal coordination; can oversee and authorise AI decisions; can handle several concurrent emergencies.
Ambulance and emergency-response services (for example 1122 dispatchers and crews)	Request and follow emergency responses	Faster, traffic-aware routes and hospital selection; shorter travel and waiting time; fewer stops at intersections.
Patients and their families	Indirect beneficiaries	Shorter response times for time-critical cases such as cardiac arrest, stroke and major trauma.
Hospitals	Indirect beneficiaries	More predictable ambulance arrival times to prepare for incoming patients.
General road users	Affected by signal changes	Corridor design limits extra delay, queue growth and recovery time for normal traffic.
Researchers and students	Read-only research users	Reproducible SUMO digital twin and open evaluation setup for further traffic and AI research.

12. Data Access, Security and Compliance

If PSCA collaboration is confirmed, the team will first produce a technical requirements document that settles what is available and permitted. Until then, the project proceeds with public, historical or synthetic data.

The requirements document will settle:

- What traffic, camera, signal, GIS and emergency data are available.
- Whether raw CCTV or processed detections are provided, and whether historical data can be used.
- What GPU, software and network resources exist on PSCA systems.
- Whether external APIs and AI coding tools are permitted.
- Whether models or code may leave the PSCA environment.
- What level of signal-system integration is authorised.

Security principles:

- All work involving PSCA data is carried out on PSCA-provided systems, in line with PSCA's data-security policy; no data is copied to personal devices.
- Sensitive data (camera frames, internal network information, raw surveillance data) is processed locally and never sent to external services.
- Emergency vehicles are identified by anonymous IDs; only location, speed, heading, timestamp and destination are needed.
- The system is offline-first, uses role-based access (Admin, PSCA Operator, Traffic Operator, Research read-only) and keeps audit logs.
- AI is advisory: all commands pass a safety validator and, in live use, operator authorisation.

13. Cloud Platform and Deployment Approach

The project does not plan to use a public cloud platform; it plans an on-premise, offline-first deployment on Linux (Ubuntu). This is a tentative design choice and is subject to PSCA requirements and to university and advisor guidance.

- **Why on-premise.** PSCA's environment is expected to be restricted and offline-first, and sensitive camera and traffic data must not leave local systems. Cloud-only services are therefore treated as incompatible with the target environment.
- **What runs locally.** Python services (FastAPI), PostgreSQL + PostGIS, Redis, SUMO, model training and inference, and the React dashboard served as static files behind Nginx with TLS.

- **During development.** The SUMO digital twin and models are built and tested on team or university machines using public (OpenStreetMap), historical or synthetic data until PSCA access is confirmed. No sensitive PSCA data is used outside PSCA-approved systems.
- **If requirements change.** If PSCA or the university later requires or permits a specific cloud or hosting platform, the modular, adapter-based design allows the deployment target to change without redesigning the AI components. Optional tools such as Docker will be used only if permitted.

14. External Collaboration and Funding Status

No external collaboration or funding is confirmed at this time. Everything related to PSCA, including its data, systems and a technical mentor, is proposed but not yet agreed.

Item	Current status
University supervision	Confirmed: Prof. Hafiz Ahsan Arshad (advisor) and Dr. Muhammad Husnain Ashfaq (co-advisor), UMT.
PSCA collaboration	Not confirmed. PSCA / PPIC3 is the proposed environment only.
PSCA technical mentor	Not confirmed.
Access to PSCA data (CCTV, AVLS, PTSC, GIS)	Not requested or not granted; any access would be subject to PSCA authorisation.
External funding	None confirmed. The project is planned to run on open-source tools and existing university resources.

The project is designed to succeed without PSCA: the SUMO digital twin (Mode 2) is the main development and evaluation environment, and PSCA integration (Mode 1) is an optional extension. If collaboration is later confirmed, this section will be updated with its scope and terms.

15. Tools and Technology

Tools were chosen to be open-source, able to work offline in a restricted environment, proven in research and industry, explainable and testable, and learnable within a final-year project. Python is the common language for CV, ML, RL, simulation, GIS and the API.

Area	Technologies	Role
Frontend	React + TypeScript, Vite, MapLibre GL JS, Tailwind CSS, Apache ECharts, WebSocket	Operator dashboard with offline-capable live map, charts and real-time updates.
Backend and services	Python 3.11, FastAPI + Uvicorn, Pydantic, SQLAlchemy + GeoAlchemy2 + Alembic, Redis, JWT + role-based access, Nginx	REST and WebSocket API, strict data validation, live state cache, authentication and on-premise TLS.
Data, GIS and database	PostgreSQL + PostGIS, OpenStreetMap + OSMnx / NetworkX, GeoPandas, Shapely, PyProj, Pandas, NumPy, PyArrow / Parquet	Spatial queries, road graph, map matching, ETL and feature engineering.
Computer vision	OpenCV, Ultralytics YOLO (PyTorch), ByteTrack / BoT-SORT, homography calibration	Vehicle detection, tracking, speed and queue measurement.
Machine learning	scikit-learn, XGBoost / LightGBM, PyTorch (LSTM / GRU)	Baselines, ETA and traffic prediction, sequence forecasting.
Simulation, routing, optimisation, RL	SUMO, TraCI, libsumo, netedit / netconvert / duarouter, NetworkX (Dijkstra, A*), Google OR-Tools, Gymnasium + Stable-Baselines3 (PPO)	Digital twin, routing, constrained signal optimisation and RL training.
Agentic AI	LangGraph, local / approved LLM (for example via Ollama or vLLM)	Emergency Manager with Traffic, Route and Signal agents; plans and explanations only.
Engineering and MLOps	Git, MLflow (+ DVC optional), Docker (if permitted), pytest, Ruff, Black, MyPy, MkDocs, Postman, Linux (Ubuntu) on-premise	Version control, experiment tracking, testing, documentation and runtime.

Tools deliberately not chosen

- Kafka, Kubernetes, microservices – unnecessary complexity; a modular monolith with Redis is enough.
- MongoDB – a poor fit for spatial, relational road-network queries compared with PostGIS.
- Transformer traffic models (at first) – data-hungry; XGBoost and LSTM are tested first.
- LLM-controlled signals – unsafe; LLMs only plan and explain.
- Cloud-only services – incompatible with a restricted, offline-first environment (Section 13).

AI-assisted development

AI coding assistants may be used to speed up code, tests and documentation, only if permitted by the university and, where relevant, PSCA, and never with raw PSCA surveillance data, camera frames or internal network details. The team reviews and understands every component.

16. Expected Outcomes and Deliverables

The target result is a measurable reduction in ambulance travel and waiting time versus the baselines, without disproportionate delay to normal traffic, in a system ready to connect to PSCA's infrastructure if and when authorised.

- A calibrated SUMO digital twin of a Lahore corridor.
- A traffic-analytics pipeline (detection, tracking, speed, density, congestion).
- Trained ETA and traffic-prediction models with evaluation results.
- Traffic-aware routing with dynamic re-routing, and a predictive multi-intersection green-corridor engine with a safety layer.
- An RL controller, a constrained agentic coordination layer and a React operator dashboard.
- A comparative evaluation of six strategies, a final report / thesis and a live demonstration.

Summary

Traffic and ambulance data → digital twin → AI prediction → dynamic route optimisation → green-corridor decision → safety validation → traffic-signal coordination. The system is proven first in SUMO and connects to PSCA's infrastructure only where access is authorised.